

ANDREWS AFB, MD, USA

WMO: 745940

Lat: 38.817N

Lon: 76.867W

Elev: 86

StdP: 100.30

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13705

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.0	(c) -7.7	(d) -19.1	(e) 0.7	(f) -8.1	(g) -16.3	(h) 0.9	(i) -5.4	(j) 12.7	(k) -0.2	(l) 11.8	(m) 0.1	(n) 4.6	(o) 320	(p) 0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	34.3	23.8	32.8	23.4	31.4	22.8	25.6	31.4	24.9	30.2	24.1	29.0	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.1	27.6	23.2	18.2	26.9	22.7	17.6	26.4	78.6	31.3	75.6	30.2	72.9	29.0	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.3	8.3	-13.8	36.1	3.0	1.6	-15.9	37.3	-17.7	38.2	-19.3	39.1	-21.5	40.3		
			-14.4	27.1	2.7	1.4	-16.4	28.1	-17.9	28.9	-19.5	29.7	-21.4	30.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	1.4	3.1	7.3	13.3	18.2	22.9	25.3	24.3	20.7	14.5	8.5	4.1		
	DBStd	9.44	5.90	5.43	5.42	4.88	4.33	3.35	2.81	2.73	3.60	4.48	4.65	5.03		
	HDD10.0	904	277	202	118	19	1	0	0	0	0	10	82	194		
	HDD18.3	2416	527	426	343	166	58	6	1	0	17	137	296	441		
	CDD10.0	2254	8	9	36	119	254	388	476	444	322	148	38	12		
	CDD18.3	722	0	0	2	14	54	143	217	186	88	17	1	0		
	CDH23.3	6467	0	1	20	163	491	1271	2131	1611	647	124	6	1		
	CDH26.7	2428	0	0	3	45	151	463	926	612	202	27	0	0		
(14) Wind	WSAvg	3.7	4.4	4.4	4.6	4.3	3.6	3.4	3.1	2.9	3.2	3.4	3.8	3.9		
(15) Precipitation	PrecAvg	1132	80	67	98	91	106	115	109	102	100	91	70	92		
	PrecMax	1645	156	180	225	151	255	273	180	256	277	217	157	168		
	PrecMin	889	37	12	0	49	32	34	30	37	2	7	10	37		
	PrecStd	162	29	36	44	30	53	60	41	49	66	53	41	34		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	21.6	26.0	31.0	32.5	34.8	36.7	36.0	33.6	29.5	24.0	20.8		
		MCWB	15.8	14.8	15.9	19.8	22.7	24.1	24.7	23.9	22.4	20.8	17.2	16.6		
	2%	DB	17.0	17.7	22.4	27.7	30.3	32.7	34.9	33.5	31.3	26.7	20.9	17.6		
		MCWB	14.1	12.6	14.5	17.6	21.7	23.5	24.3	23.8	22.1	19.9	15.7	14.4		
	5%	DB	13.9	14.5	19.4	24.9	28.2	31.2	33.0	31.7	28.9	24.1	18.6	14.9		
		MCWB	10.7	9.5	13.1	16.8	20.6	22.7	23.7	23.1	21.2	18.5	14.3	11.8		
	10%	DB	10.2	11.6	16.5	22.2	26.2	29.6	31.5	30.1	27.0	22.1	16.3	12.1		
		MCWB	7.1	7.6	11.4	15.3	19.4	22.1	23.2	22.5	20.6	17.6	12.3	9.1		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	16.6	17.8	21.0	24.2	25.7	27.0	26.3	24.9	23.0	19.5	17.9		
		MCDB	18.5	20.6	22.9	26.4	30.0	31.7	33.3	32.1	29.3	26.3	21.5	19.4		
	2%	WB	14.5	13.5	15.8	19.4	22.9	24.7	25.8	25.3	23.9	21.4	17.4	15.2		
		MCDB	16.8	16.8	20.6	25.1	28.3	30.4	32.0	30.7	28.0	24.5	19.3	17.3		
	5%	WB	11.1	10.5	14.1	17.9	21.8	24.0	25.1	24.5	23.1	20.1	15.3	12.5		
		MCDB	13.4	13.2	18.3	23.2	26.9	29.2	30.7	29.2	26.6	23.2	17.5	14.4		
	10%	WB	7.6	7.9	11.9	16.3	20.6	23.2	24.3	23.7	22.2	18.2	13.3	9.4		
		MCDB	9.4	11.1	16.1	20.9	24.5	27.8	29.4	27.9	25.4	21.2	15.6	11.4		
(36) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.6	10.5	11.6	10.9	10.5	10.3	10.2	10.1	10.6	10.1	8.8		
		MCDBR	12.5	13.7	14.6	15.1	13.4	12.3	12.1	11.9	12.4	12.8	12.5	11.9		
	5% WB	MCWBR	9.8	9.5	9.0	7.5	6.4	5.1	4.3	4.4	5.1	6.7	8.3	9.4		
		MCWBR	11.7	11.7	13.1	13.1	11.9	11.0	10.6	10.0	9.6	10.3	10.0	10.5		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.318	0.352	0.394	0.434	0.488	0.511	0.504	0.418	0.357	0.321	0.307		
	taud		2.501	2.450	2.415	2.314	2.231	2.122	2.076	2.079	2.294	2.470	2.530	2.534		
	Ebn at Noon		875	909	907	883	851	802	779	773	825	852	849	848		
	Edh at Noon		78	93	107	126	139	156	162	157	120	91	75	71		
(44) All-Sky Solar Radiation	RadAvg		2.10	2.94	3.92	4.96	5.59	6.18	6.06	5.40	4.39	3.30	2.39	1.85		
	RadStd		0.17	0.33	0.29	0.37	0.64	0.47	0.30	0.29	0.50	0.36	0.20	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (78 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+76

Nomenclature: See separate page